Python-based codes for WDM-transform for use within gravitational wave astronomy

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Here we present `PyWavelet`, a Python package for rapid and accurate Wilson-Daubechies-Meyer (WDM) time-frequency transforms on CPU and accelerator backends. The current implementations use NumPy and JAX and can transform million-point data streams in milliseconds while maintaining excellent numerical accuracy. We give a practical introduction to the WDM domain, discuss several implementation choices and limitations, and illustrate how these methods can support future space-based gravitational-wave data analysis.

I. INTRODUCTION

Time-frequency analysis is widely used in gravitational-wave astronomy. It enables template-free modeling of source signals [4, 8], instrumental transients [7], and the joint analysis of astrophysical signals and terrestrial disturbances [2]. Moreover, time-frequency representations provide compact waveform descriptions and facilitate the modeling of non-stationary detector noise. Among the available techniques, discrete wavelet wavepacket transforms are particularly well suited to gravitational-wave data analysis: they furnish localized, orthonormal basis functions capable of representing transient and chirping signals, and for locally stationary noise they yield an approximately diagonal noise covariance, greatly simplifying likelihood evaluation and spectral estimation.

The Wilson–Daubechies–Meyer (WDM) wavelet wavepacket basis [6] was introduced into gravitational-wave analysis through the coherent WaveBurst algorithm [**?** **?**] for unmodeled burst searches. Building on this foundation, Cornish [1] developed WDM into a general-purpose framework for template-based inference, demonstrating that binary merger waveforms can be generated directly in the WDM domain at a computational cost scaling as $N\sqrt{N}$ rather than N^2, and showing that the WDM noise covariance is diagonal for a broad class of locally stationary processes. These results provide strong motivation for performing gravitational-wave inference directly in the WDM domain, particularly for long-duration signals expected in future detectors such as LISA.

While the foundational works above establish the theoretical and algorithmic advantages of the WDM transform, many practical aspects of its implementation are necessarily distributed across the literature and existing codebases. In particular, details such as the packed coefficient layout, the treatment of the DC and Nyquist edge channels, and the numerical properties of the discretized transform are often implicit. As a result, adopting WDM in new applications can require reconstructing these elements from multiple sources.

This paper aims to provide a unified and pedagogical presentation of the WDM transform. We describe what each coefficient represents, how the forward and inverse transforms are computed step by step, how the packed (N_t, N_f+1) coefficient grid is organized in practice, and under what conditions the coefficients may be treated as approximately independent. We accompany this exposition with `PyWavelet`, an open-source Python package that implements the transform on both CPU (NumPy) and accelerator (JAX) backends, with validation through round-trip reconstruction tests and empirical decorrelation studies.

The motivation for WDM can be understood by comparing it with the two standard representations of a discrete signal. In the time domain one can identify *when* a feature occurs but not which frequency band is active, while in the Fourier domain one can identify *which* frequencies are present but not when they appear. Many gravitational-wave signals and detector artifacts are neither purely stationary nor purely impulsive, so neither representation is fully satisfactory in isolation. A short-time Fourier transform partially addresses this limitation through a sliding window, but imposes a fixed time-frequency resolution across the dataset. The WDM transform instead tiles the time-frequency plane with localized atoms indexed by a time bin n and a frequency channel m, with coefficients w_{nm} measuring the projection of the data onto each atom. The result is a uniform time-frequency representation that is well suited to describing transients, chirps, gaps, and other localized structure, while still admitting an

exact inverse transform back to the original sampled data.

The remainder of this paper is organized as follows. Section I A introduces notation and conventions. Section II develops the WDM transform, including the Meyer window, the full three-case basis definition covering DC, interior, and Nyquist channels, the forward and inverse transforms, and the statistical properties of the coefficients under stationary and non-stationary noise. Section III describes the `PyWavelet` package, presents runtime and accuracy benchmarks, and demonstrates the transform on astrophysical signals. We conclude in Section IV..

A. Conventions

We are given a data stream in time domain, consisting of a set of N samples of a **real** quantity x at times $\{t_k\}_{k=0\ldots N-1}$ in a total time of observation T

$$x[k] = x(t_k) \qquad , \qquad k = \{0 \ldots N-1\} \ , \tag{1}$$

sampled uniformly with sampling interval $\Delta t = t_{k+1} - t_k$ with $T = N\Delta t$. All of the information we have lies in these N points. We define the time sampling rate and the Nyquist frequency as

$$\begin{aligned} r_s &\equiv \frac{1}{\Delta t} \equiv \frac{N}{T} \ , \\ f_{\max} &= \frac{1}{2} r_s \ , \\ \Delta f &= 1/T \end{aligned} \tag{2}$$

With that respect, if we set $t_0 = 0$, we can write $t_k = k\Delta t$. In the frequency domain, the data stream $\{x_k\}$ can be discrete-Fourier-transformed to a set of other N complex numbers as

$$\tilde{x}[l] = \sum_{k=0}^{N-1} x[k] e^{-2\pi i l k/N} \tag{3}$$

for Fourier frequencies $f_l = \{-N\Delta f/2, \ldots, -\Delta f, 0, \Delta f, (N/2 - 1)\Delta f\}$. This is convenient since the zeroth frequency bin has been shifted into the middle of the spectrum.

B. Time and Frequency Domains

We assume that the output of a gravitational wave detector, $x(t)$, is expressed as a linear combination of a signal, $h(t|\vec{\mu})$, determined by a finite set of (unknown) parameters, $\vec{\mu}$, and instrumental noise, $n(t)$. If we ignore the presence of calibration errors [**?**], the content of a single data stream, i.e., one output channel from one detector, can be written in the time and frequency domain as:

$$x[k] = h[k|\vec{\mu}] + n[t], \tag{4}$$

$$\tilde{x}[l] = \tilde{h}[l|\vec{\mu}] + \tilde{n}[l] \tag{5}$$

where the tilde indicates the Fourier transform.

In a gravitational wave context it is usual to further assume that the instrumental noise follows a Gaussian distribution characterized by a one-sided PSD, $S_n(f)$, defined such that

$$\langle \tilde{n}^*[l] \tilde{n}[l'] \rangle = \frac{1}{2} P[l] \delta_{ll'}, \tag{6}$$

where the expectation value $\langle\,\rangle$ is taken over the data generating process. The delta function in the previous equation implies that different frequencies are not correlated.

In reality the noise model is not known perfectly and could vary from the assumption above in a number of ways. For example, the PSD might have a different shape from the reference one [**?**], the probability distribution of the noise might not be Gaussian, or the noise might not be stationary, leading to correlations between frequencies.

II. THE WILSON-DAUBECHIES-MEYER (WDM) TRANSFORM

Before defining the transform, it is worth clarifying a point of terminology. A standard discrete wavelet transform (DWT) decomposes a signal using a dyadic tiling of the time-frequency plane: each successive scale doubles the frequency bandwidth and halves the time resolution. This is natural for self-similar or fractal-like signals, but it means that low-frequency channels have coarse time resolution and high-frequency channels have coarse frequency resolution. A *wavelet packet* (or *wavelet wavepacket*) transform generalizes this by allowing the user to choose any admissible tiling of the time-frequency plane. The WDM basis is a specific wavelet packet basis that uses a *uniform* rectangular tiling: every cell has the same width ΔT in time and the same width ΔF in frequency, with $\Delta T\,\Delta F = 1/2$. This uniform tiling is what makes the WDM representation so well suited for gravitational-wave data analysis, because signals that sweep across many frequency bands (such as compact binary inspirals) occupy a simple, coherent track on the grid rather than being spread across cells of very different sizes. Throughout this paper, we use "WDM transform" and "wavelet transform" interchangeably, but the reader should keep in mind that the WDM basis belongs to the wavelet packet family rather than to the standard dyadic wavelet family.

The WDM transform maps the original N samples to a packed coefficient grid with N_t time bins and $N_f + 1$ stored frequency channels. The transform size is still set by

$$N = N_t\, N_f \ , \tag{7}$$

and the stored channels are indexed by $m = 0, \ldots, N_f$, with $m = 0$ the DC edge channel, $m = 1, \ldots, N_f - 1$ the interior channels, and $m = N_f$ the Nyquist edge channel. The underlying idea is to chunk the data stream into N_t chunks and to perform a frequency analysis of each one of those. Please note that N is fixed, but N_t is an arbitrary choice you can make. The only constraint is obviously $1 \leq N_t \leq N$. Consequently, the constraint applies for N_f: $1 \leq N_f \leq N$. Given that choice we can define the spacing in both time and frequency with respect to the underlying tiling with area $N = N_t N_f$

$$\begin{aligned} \Delta T &\equiv \frac{T_{\rm obs}}{N_t} \qquad \text{grid time spacing,} \\ \Delta F &\equiv \frac{f_{\rm max}}{N_f} \qquad \text{grid frequency spacing}\,. \end{aligned} \tag{8}$$

One immediately sees that,

$$\Delta T = \frac{T_{\rm obs}}{N_t} = \frac{N\Delta t}{N_t} = N_f \Delta t \tag{9}$$

$$\Delta F = \frac{f_{\rm max}}{N_f} = \frac{1}{2N_f\Delta t} = \frac{N_t \Delta f}{2} = \frac{1}{2\Delta T} \tag{10}$$

This final expression shows how the Nyquist theorem still holds. Once you fix N_t (or equivalently N_f, ΔT or ΔF) the grid is uniquely identified. Figure 1 summarizes the packed layout, including the DC and Nyquist edge channels and the alternating phase factor used for the interior coefficients.

A. Forward transform

For a given frequency series $\tilde{x}[l]$, the WDM transform represents the data through the coefficients

$$\begin{aligned} w_{nm} &= \sum_{l=0}^{N-1} \tilde{x}[l]\tilde{g}^*_{nm}[l] \ , \\ \tilde{x}[l] &= \sum_{n=0}^{N_t}\sum_{m=0}^{N_f} w_{nm}\tilde{g}_{nm}[l] \ , \end{aligned} \tag{11}$$

where n corresponds to the time index and m corresponds to the frequency (or scale) index. The functions $g_{nm}[k]$ are the WDM basis elements, which are chosen to be (quasi-)orthonormal:

$$\begin{aligned} \sum_{l=0}^{N-1} g^*_{nm}[l] g_{pq}[l] &= \delta_{(nm),(pq)} \ , \\ \sum_{n=0}^{N_t-1}\sum_{m=0}^{N_f} g^*_{nm}[l] g_{nm}[l'] &= \delta_{ll'} \ , \end{aligned} \tag{12}$$

This orthonormality is also the key reason that WDM coefficients can become statistically simple for stationary noise. If the Fourier-domain noise satisfies Eq. (6), then the covariance between two WDM noise coefficients is

$$\langle w_{nm} w_{pq}\rangle = \frac{1}{2}\sum_l P[l]\, \tilde{g}_{nm}(l)\tilde{g}^*_{pq}(l). \tag{13}$$

In the ideal orthonormal limit, whitened noise, or more generally noise with $P[l]$ effectively constant across the support of each atom, reduces this covariance to a diagonal form. In that case the WDM coefficients are uncorrelated, and for Gaussian noise they are therefore independent. In the sampled packed implementation used here, it is useful to separate two statements. First, the interior atoms ($m = 1, \ldots, N_f - 1$) satisfy the discrete orthonormality relation numerically to machine precision. Second, the packed real-valued storage includes special DC and Nyquist edge channels, so the full stored coefficient array is best viewed as a convenient packed representation rather than as a minimal orthonormal basis. For this reason, the decorrelation discussion below focuses on the interior coefficients. For whitened noise their empirical correlations remain very small, while for colored, non-stationary, or non-Gaussian noise off-diagonal structure becomes more visible and the coefficients should be regarded as only approximately decorrelated rather than strictly independent. Figure 2 illustrates these points numerically for a small sampled transform and for white and colored noise realizations.

The basis elements are defined in closed form in the frequency domain. For interior channels ($0 < m < N_f$), DC ($m = 0$), and Nyquist ($m = N_f$) the definitions differ:

$$\tilde{g}_{nm}(f) = \frac{1}{\sqrt{2}} \begin{cases} e^{-4\pi i n f \Delta T}\, \tilde{\varphi}(f), & m = 0, \\ e^{-2\pi i n f \Delta T}\Big(C_{nm}\, \tilde{\varphi}(f - m\Delta F) & \\ \quad + C^*_{nm}\, \tilde{\varphi}(f + m\Delta F)\Big), & 0 < m < N_f, \\ e^{-4\pi i n f \Delta T}\Big(\tilde{\varphi}(f - N_f\Delta F) & \\ \quad + \tilde{\varphi}(f + N_f\Delta F)\Big), & m = N_f, \end{cases} \tag{14}$$

for $n = 0, \ldots, N_t - 1$. The interior channels carry a $1/\sqrt{2}$ normalization and the alternating phase factor C_{nm},

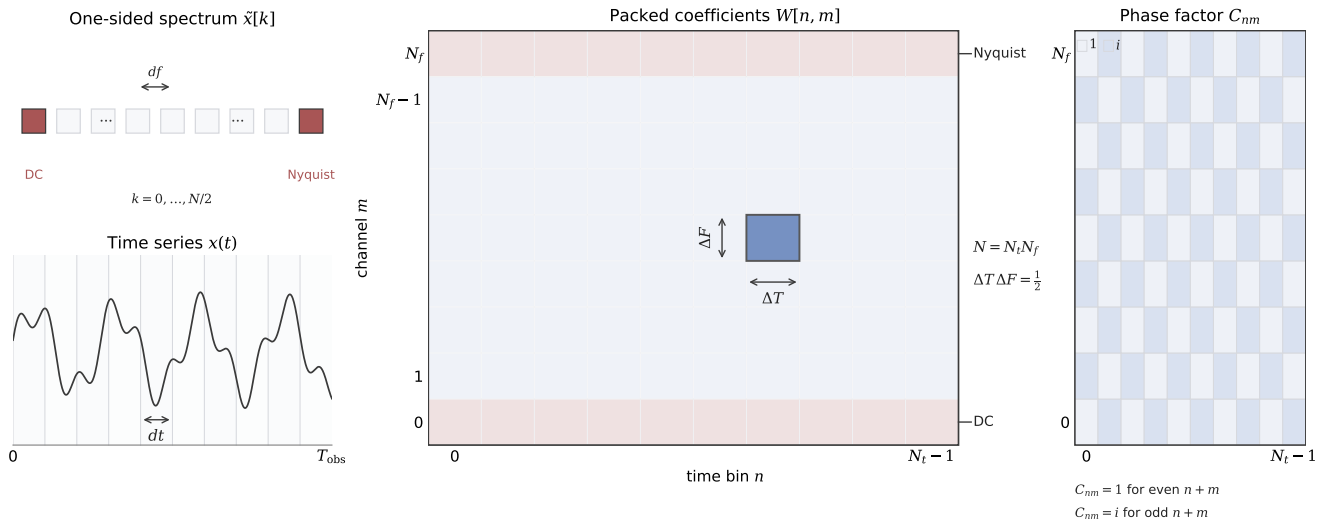

Figure 1. **Packed WDM coefficient layout.** The left panels sketch the same data in the one-sided frequency series and in the time domain, emphasizing the frequency index range $k = 0, \ldots, N/2$ and the time-bin spacing $\Delta T = T_{\rm obs}/N_t$. The center panel shows the packed WDM coefficient array $W[n, m]$, whose stored channel index runs from $m = 0$ to $m = N_f$, so the array has shape $N_t \times (N_f + 1)$ even though the transform size is fixed by $N = N_t N_f$. The right panel shows the alternating phase factor C_{nm}, which equals 1 when $n + m$ is even and i when $n + m$ is odd.

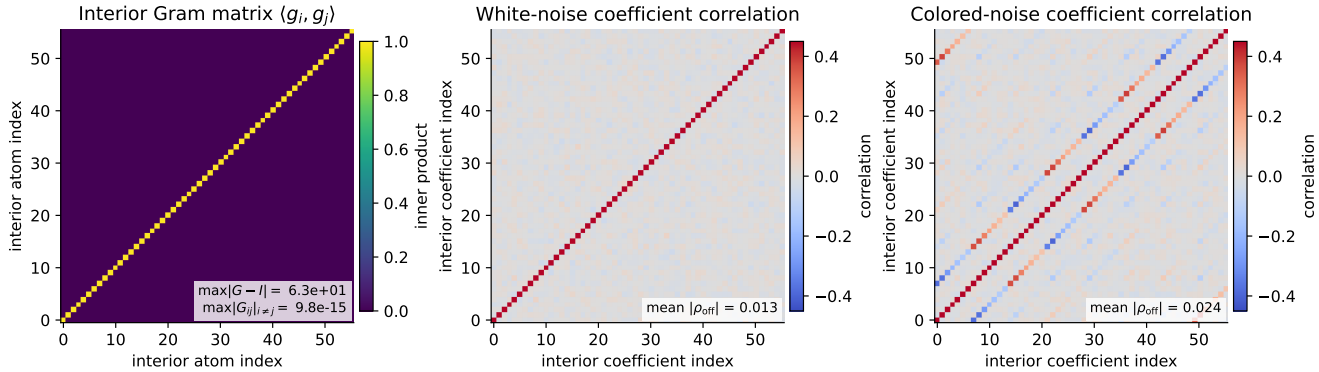

Figure 2. **Numerical checks of orthogonality and decorrelation.** Left: Gram matrix of the interior sampled WDM atoms, showing the discrete inner products $\langle g_i, g_j \rangle$ after flattening the interior index pair (n, m) into a single label i. The matrix is visually indistinguishable from the identity, and the annotated residuals are at machine precision, providing a direct numerical demonstration of orthonormality for the interior channels. Middle: empirical correlation matrix of interior WDM coefficients estimated from many realizations of white Gaussian noise; most off-diagonal entries remain small. Right: the same calculation for colored stationary noise, where visibly stronger off-diagonal correlations appear.

while the DC and Nyquist edge channels have a doubled time-shift exponent and no C_{nm} modulation. These edge channels each contribute N_t real degrees of freedom to the packed grid, accounting for the $N_t(N_f + 1)$ stored entries versus the $N = N_t N_f$ independent samples.

The window $\tilde{\varphi}$ is the Meyer window, defined in the frequency domain by

$$\tilde{\varphi}(\omega) = \begin{cases} \frac{1}{\sqrt{\Delta\Omega}} & |\omega| < A \\ \frac{1}{\sqrt{\Delta\Omega}} \cos\left[\frac{\pi}{2}\left(\frac{|\omega| - A}{B}\right)\right] & A \leq |\omega| \leq A + B \\ 0 & \text{otherwise,} \end{cases} \tag{15}$$

where $\omega = 2\pi f$ are angular frequencies, which are dis-

cretized as

$$\omega_l = \frac{2\pi l}{N\Delta t} \tag{16}$$

By construction, $\tilde{\varphi}$ has a limited frequency bandwidth which is controlled by the parameter A. In this work, we restrict attention to the $d = 1$ case, for which the Meyer window reduces to the cosine taper written above; this is also the case implemented in the current package, while d is retained in the API only for possible future generalizations.[1] The parameter $\Delta\Omega$ depends on the frequency resolution

$$\Delta\Omega = 2\pi\Delta\tilde{F}, \tag{17}$$

where $\Delta\tilde{F} = \Delta F\Delta t$ is the normalized frequency resolution. The parameter B depends on $\Delta\Omega$ and A as

$$B = \Delta\Omega - 2A \tag{18}$$

In our implementation, we set $A = \Delta\Omega/4$ and therefore $B = \Delta\Omega/2$. This means that the window is non-zero when $|\omega| \leq 3\Delta\Omega/4$. Figure 3 summarizes the window shape, its shifted channel copies, and the localized atoms that result after modulation in time.

These are the main ingredients behind the packed WDM grid. A single window shape is shifted across frequency to define the channel index m, and each shifted window is then modulated in time to define the atom indexed by n. This is why one WDM coefficient can be read as the strength of one localized time-frequency pattern, rather than as the amplitude of a globally supported sinusoid.

More concretely, a single coefficient w_{nm} answers the question: "how much does the data resemble the atom centered at time bin n and frequency channel m?" A large magnitude means that the data contain a feature with roughly the same time localization, central frequency, and oscillatory phase structure as that atom. Reading the WDM plane coefficient-by-coefficient in this way is often much more intuitive than interpreting a global Fourier series, because the index pair (n, m) already carries an approximate time-frequency location.

We now derive the practical computational form of the forward WDM coefficients by inserting the basis functions (14) into the definition (11). The phase factor C_{nm} alternates between real and imaginary depending on $n + m$:

$$C_{nm} = e^{\frac{i\pi}{4}(1-(-1)^{n+m})} = \begin{cases} 1 & n+m \text{ even}, \\ i & n+m \text{ odd}. \end{cases} \tag{19}$$

We treat the three channel types separately.

1. *Interior channels* ($0 < m < N_f$)

Starting from Eq. (11) with the interior-channel basis element from Eq. (14), the forward transform can be evaluated in the Fourier domain using

$$w_{nm} = \sum_{l=0}^{N-1} \tilde{x}_l\, \tilde{g}^*_{nm}(f[l])\,, \tag{20}$$

where $\tilde{x}_l = X[l]$ is the DFT of the input data. For a real-valued input signal, $\tilde{x}_{N-l} = \tilde{x}^*_l$, and the window satisfies $\tilde{\varphi}(-f) = \tilde{\varphi}(f)$. Splitting the sum into positive and negative frequency halves and using these symmetries, the negative-frequency contribution is the complex conjugate of the positive-frequency contribution (see Appendix B for details), giving

$$w_{nm} = \sqrt{2}\,(-1)^{nm}\,\Re\left[C^*_{nm} \sum_{l'=-N_t/2}^{N_t/2} \tilde{x}_{l'+mN_t/2}\, e^{2\pi i l' n/N_t}\, \tilde{\varphi}(l'\,df)\right], \tag{21}$$

where $l' = l - mN_t/2$ and $df = 1/T$. This can be written compactly as

$$w_{nm} = \sqrt{2}\,(-1)^{nm}\,\Re\, C^*_{nm}\, x_m[n]\,, \tag{22}$$

where

$$x_m[n] = \sum_{l=-N_t/2}^{N_t/2-1} e^{2\pi i l n/N_t}\, X[l + mN_t/2]\, \tilde{\varphi}[l] \tag{23}$$

is the windowed, shifted inverse DFT of length N_t. The quantity $x_m[n]$ can be computed using a length-N_t FFT for each channel m, so the total cost of the interior channels is $N_f \times \mathcal{O}(N_t \log N_t)$.

2. *DC channel* ($m = 0$)

The DC edge channel has a doubled time-shift exponent and no C_{nm} factor in the basis definition (14). Following the same Fourier-domain evaluation yields

$$w_{n0} = \sqrt{2}\,\Re\left[\sum_{l=1}^{N_t/2} \tilde{x}_l\, e^{4\pi i l n/N_t}\, \tilde{\varphi}(l\,df)\right] + \frac{1}{\sqrt{2}}\tilde{x}_0\, \tilde{\varphi}(0)\,. \tag{24}$$

[1] Cornish [1] uses $d = 4$, which yields a flatter passband and steeper roll-off at the expense of a wider time-domain support (the window function must be truncated at $\pm q\Delta T$ with $q = 16$). The $d = 1$ cosine taper used here has a gentler roll-off but is more compact in time, requiring a smaller q. In practice, the choice of d trades spectral leakage suppression against temporal compactness of the window; for the applications considered in this paper the $d = 1$ window provides adequate frequency separation while keeping the implementation simple and the time-domain filter short.

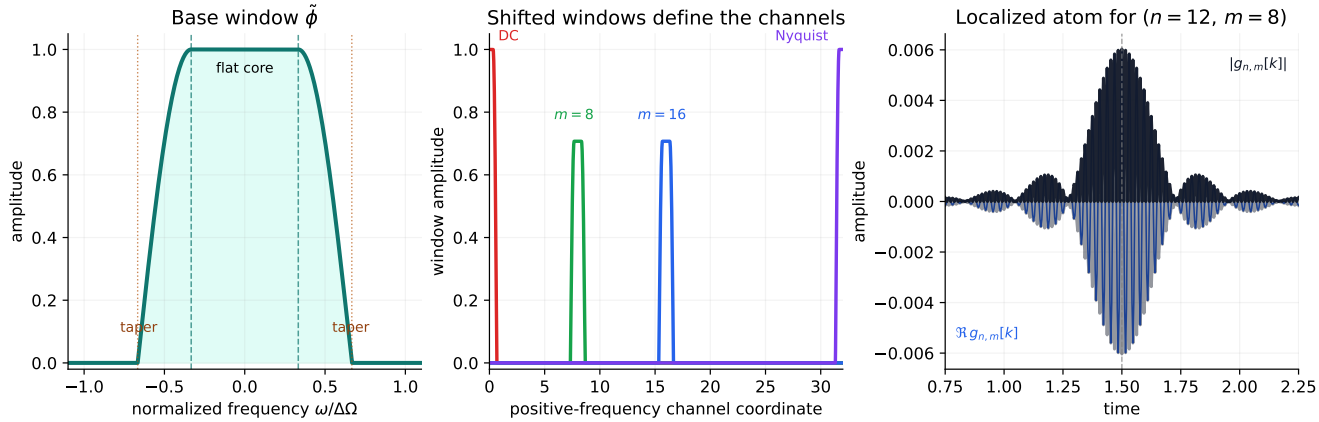

Figure 3. **Pedagogical view of the WDM window and atoms.** Left: the cosine-tapered Meyer window used in the current implementation. Middle: shifted copies of the same window define the WDM frequency channels, with special edge-channel behavior at DC and Nyquist. Right: after fixing a channel m and time index n, the corresponding inverse-Fourier-transformed atom becomes a localized oscillatory pattern in time. This is the object that one WDM coefficient measures against the data.

Note the factor of 4π in the exponent (versus 2π for interior channels): the DC basis element oscillates at twice the rate in the time index because it sees both positive and negative frequency copies of the window centered at $f = 0$.

3. *Nyquist channel ($m = N_f$)*

The Nyquist edge channel is handled analogously to the DC channel but centered at the Nyquist frequency $f_{\max}$. Its structure mirrors the DC case with appropriate frequency shifts. In practice, the Nyquist channel is computed using the same windowed-FFT machinery as the interior channels, with the frequency shift set to $mN_t/2 = N/2$.

$$w_{nN_f} = \sqrt{2}\,\Re\left[\sum_{l=N/2-N_t/2}^{N/2-1} \tilde{x}_l\, e^{4\pi i l n/N_t}\, \tilde{\varphi}((l - N/2)\, df)\right] + \frac{1}{\sqrt{2}} \tilde{x}_{N/2}\, \tilde{\varphi}(0)\,. \tag{25}$$

B. Backward transform

The original data stream is recovered by applying the backward (inverse) WDM transform to the coefficients:

$$x[k] = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} w_{nm}\, g_{nm}[k]\,. \tag{26}$$

Equivalently, in the Fourier domain the reconstruction reads

$$\tilde{x}_l = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} \tilde{g}_{nm}[l]\, w_{nm}\,. \tag{27}$$

That this is indeed the inverse of the forward transform follows from the (quasi-)orthonormality of the basis: inserting the forward transform (11) into (27) and using (12) returns $\tilde{x}_l$ up to the overlap of adjacent window functions. Specifically, the sum over n produces $N_t \delta_{ll'}$ factors via the discrete orthogonality of the complex exponentials, and the sum over m covers the full frequency axis because adjacent windows satisfy $\tilde{\varphi}^2(\omega) + \tilde{\varphi}^2(\omega - \Delta\Omega) = 1/\Delta\Omega$ in their overlap region (proven in Appendix A 1). Together these two identities ensure exact round-trip reconstruction: forward followed by inverse returns the original data to machine precision, as verified numerically in Section III.

C. Statistical properties and the WDM likelihood

The practical payoff of the (approximate) orthonormality and decorrelation discussed above is that standard gravitational-wave data analysis quantities — the likelihood, signal-to-noise ratio, and Fisher information matrix — can all be written as simple sums over WDM pixels, in direct analogy with the familiar frequency-domain expressions.

a. From covariance to likelihood. For Gaussian noise characterized by a covariance matrix $\mathbf{C}$, the log-

likelihood of the data $\mathbf{d}$ given a signal model $\mathbf{h}$ is

$$\ln p(\mathbf{d}|\mathbf{h}) = -\frac{1}{2}(\mathbf{d}-\mathbf{h})^\top \mathbf{C}^{-1}(\mathbf{d}-\mathbf{h}) - \frac{1}{2}\ln\det(2\pi\mathbf{C})\,. \tag{28}$$

In the Fourier domain with stationary noise, $\mathbf{C}$ is diagonal with entries proportional to the power spectral density $S(f_l)$, so the likelihood reduces to a sum over frequency bins — the Whittle likelihood. The same simplification occurs in the WDM domain whenever the noise covariance is (approximately) diagonal, which, as argued above, holds for locally stationary noise whose PSD varies slowly compared to the frequency support of each atom. In that regime the WDM noise covariance takes the form

$$\langle w_{nm}^{\rm noise}\, w_{pq}^{\rm noise}\rangle \approx S_{nm}\,\delta_{np}\,\delta_{mq}\,, \tag{29}$$

where S_{nm} is the *evolutionary power spectral density* evaluated at the time-frequency location of pixel (n, m). For stationary noise, S_{nm} reduces to the usual PSD integrated over the atom's frequency support [1]:

$$S_{nm} \approx S(f_m)\,\Delta F\,. \tag{30}$$

Substituting the diagonal covariance (29) into the general Gaussian likelihood yields the WDM Whittle log-likelihood:

$$\ln p(d|h) = -\frac{1}{2}\sum_{n,m}\left[\ln(2\pi S_{nm}) + \frac{(d_{nm}-h_{nm})^2}{S_{nm}}\right], \tag{31}$$

where d_{nm}, h_{nm} are the WDM coefficients of the data and signal model, respectively. This expression is structurally identical to the standard frequency-domain Whittle likelihood, but with the one-dimensional sum over frequency bins replaced by a two-dimensional sum over time-frequency pixels. The key advantage is that S_{nm} can vary with time, naturally accommodating non-stationary noise without requiring off-diagonal terms in the covariance.

b. Inner product, SNR, and Fisher matrix. The noise-weighted inner product in the WDM domain follows directly from (31):

$$(a\,|\,b) = \sum_{n,m}\frac{a_{nm}\,b_{nm}}{S_{nm}}\,, \tag{32}$$

from which the matched-filter signal-to-noise ratio is

$$\rho(h) = \frac{(d\,|\,h)}{\sqrt{(h\,|\,h)}}\,, \tag{33}$$

and the Fisher information matrix is

$$\Gamma_{ij} = \sum_{n,m}\frac{\partial_i h_{nm}\,\partial_j h_{nm}}{S_{nm}}\,. \tag{34}$$

These expressions assume that the diagonal approximation (29) holds. When it does not — for example, when the noise PSD varies rapidly compared to ΔT, or when the noise is strongly non-Gaussian — the off-diagonal covariance terms must be retained and the simple pixel-wise sums above become inadequate. Figure 2 provides a practical diagnostic: if the off-diagonal entries of the empirical covariance remain small after whitening, the diagonal likelihood (31) is a safe approximation.

c. When does the diagonal approximation hold? The diagonal covariance requires two conditions to be satisfied simultaneously. First, the noise must be *locally stationary*: its PSD must vary on timescales much longer than the wavelet pixel width ΔT. Second, the PSD must be approximately flat across the frequency support of each atom, which spans roughly ΔF for interior channels. The Heisenberg–Gabor uncertainty relation, saturated by the WDM tiling at $\Delta T\,\Delta F = 1/2$, means these two conditions trade off against each other. Choosing a finer frequency resolution (large ΔT) improves the second condition but worsens the first. In practice, a single choice of N_f often suffices for mildly non-stationary noise, but strongly non-stationary data with sharp spectral features may require a multi-resolution strategy, as discussed by Cornish [1].

III. PYWAVELET PACKAGE

The `PyWavelet` package provides practical implementations of the Wilson–Daubechies–Meyer (WDM) transform for gravitational-wave data analysis. The current codebase exposes a reference NumPy backend and a vectorized JAX backend; both implement the same packed (N_t, N_f+1) coefficient layout and the same forward/inverse conventions described above.

From the implementation point of view, the dominant operations in the forward transform are one length-N FFT of the input data together with one length-N_t sub-band transform for each stored channel. This gives a cost of

$$T_{\rm forward} = \mathcal{O}(N\log N) + \mathcal{O}(N\log N_t), \tag{35}$$

which is asymptotically $\mathcal{O}(N\log N)$ because $N = N_t N_f$ and $N_t \le N$. The inverse transform has the same scaling up to constant factors: it reconstructs the full spectrum from the packed channels and then applies one inverse FFT back to the time domain. In both directions, the stored coefficient grid itself contains $\mathcal{O}(N)$ real numbers, and the temporary working arrays used by the implementation are also linear in N, so the overall memory footprint is $\mathcal{O}(N)$.

In practice, the important implementation difference is not the asymptotic scaling but how the channel-wise work is executed. The NumPy backend carries out

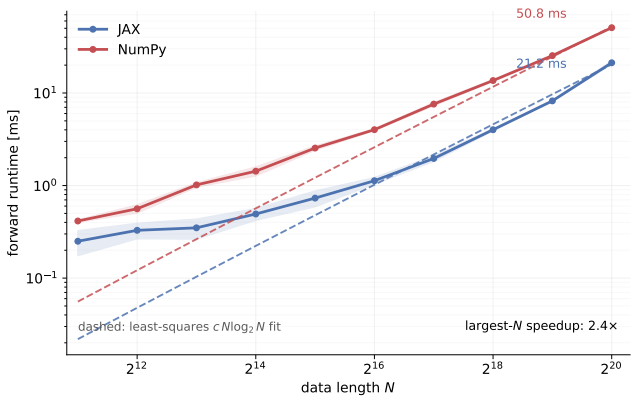

Figure 4. **Forward WDM runtime versus data length.** Each point shows the mean runtime over repeated runs after one warmup call, and the shaded band shows one standard deviation. The dashed lines show least-squares fits of the form $c\,N\log_2 N$, included as scaling guides. The JAX backend is consistently faster than the NumPy backend, with the separation becoming more pronounced for the largest transform sizes in the benchmark range.

the sub-band operations on the CPU, whereas the JAX backend evaluates the same algebra in a more vectorized form that benefits from accelerator execution. Below we use the current checked-in benchmark snapshot from the package documentation, which compares the NumPy and JAX backends across a range of input sizes. All results are obtained in 64-bit floating point precision, and each timed point uses one warmup evaluation before the measured runs. The hardware specifications for the benchmark machine are:

- **GPU:** NVIDIA [TBD] (Google Colab)
- **CPU:** [TBD]
- **RAM:** [TBD]

Figure 4 shows the forward WDM runtime as a function of data length for the NumPy (CPU) and JAX (GPU) implementations.

As visible in Figure 4, both implementations become slower as N increases, but the JAX backend maintains a noticeably lower runtime across the full benchmark range. The difference is modest for small problems and becomes much larger once the transform size reaches the upper end of the tested range. Both backends achieve round-trip reconstruction errors consistent with floating-point roundoff; the runtime differences are not due to any loss of numerical fidelity in the transform itself.

A. Resolved galactic-binary inference in a galactic background

As an end-to-end inference example, we generated a one-year LISA-like dataset containing stationary instrumental noise, a stochastic galactic foreground, and one resolved galactic binary (GB). Both the frequency-domain and WDM-domain likelihoods use the same stationary noise model throughout; this is an intentional controlled comparison designed to isolate the effect of the WDM basis change from any modelling differences. The effective PSD $S_c(f)$ in each channel c includes both the instrumental noise contribution and the time-averaged galactic foreground. Figure 5 shows the characteristic-strain comparison between the injected source and the channel noise model, and Figure 6 shows the overlaid posterior constraints from the two analyses.

The data set is constructed on a frequency grid spanning 10^{-4}–3×10^{-3} Hz. The unresolved galactic foreground is described by the phenomenological PSD [3]

$$S_{\rm gal}(f) = A_{\rm gal} f^{-7/3} \exp\left[-\left(\frac{f}{f_{1,\rm gal}}\right)^{\alpha_{\rm gal}}\right]\left[1+\tanh\left(\frac{f_{\rm knee,gal}-f}{f_{2,\rm gal}}\right)\right] \tag{36}$$

using a standard smooth fit for the LISA galactic foreground. To project this foreground into the detector channels we combine that frequency model with an anisotropic Milky Way sky map obtained by line-of-sight integration through the bulge-plus-disk density model [5]

$$\rho_{\rm gal}(x,y,z) = \rho_0\left[A e^{-r^2/R_b^2} + (1-A)e^{-u/R_d}\,\mathrm{sech}^2(z/Z_d)\right], \tag{37}$$

where $r^2 = x^2+y^2+z^2$ and $u^2 = x^2+y^2$. We evaluate the time-dependent detector response using equal-arm LISA orbits sampled at 90 orbital times, and use that response to map the anisotropic foreground into the A/E/T TDI channels, assuming that the spectrum is stationary per each of these 90 orbital times. We then draw complex Gaussian realizations of both the instrumental noise and the chunked galactic foreground and stitch them into a full-year spectrum. The resolved source is injected afterward as a TDI 1.5 A/E/T waveform with equal-arm LISA orbits and local Fourier support $n = 256$. In the realization shown here the cadence is $\Delta t = 166.7\,\mathrm{s}$, $T_{\rm obs} = 1\,\mathrm{yr}$, and the injected binary has $f_0 = 1.3862897\,\mathrm{mHz}$, $\dot f = 1.32\times10^{-18}\,\mathrm{Hz\,s^{-1}}$, $A = 1.19\times10^{-23}$, and $\phi_0 = 1.442\,\mathrm{rad}$.

The resolved GB model uses the standard eight source parameters $(f_0, \dot f, A, \alpha, \delta, \psi, \iota, \phi_0)$, but for this controlled comparison we sample only $(f_0, \dot f, A, \phi_0)$ and hold sky position and orientation fixed to their injected values. The carrier frequency is drawn near

the fixed reference value $f_{0,\mathrm{ref}} = 1.38591\,\mathrm{mHz}$ through $\delta \ln f_0 \sim \mathcal{U}[-10^{-3}, 10^{-3}]$, while $\dot{f}$ and A are drawn from truncated Gaussians in log space over $[5 \times 10^{-19}, 4 \times 10^{-18}]\,\mathrm{Hz\,s^{-1}}$ and $[6 \times 10^{-24}, 1.7 \times 10^{-23}]$, respectively.

The frequency-domain analysis uses a local A/E/T band centered on the prior support of the source and evaluates the Whittle likelihood

$$\ln \mathcal{L}_{\mathrm{freq}}(\theta) \propto - \sum_{c \in \{A,E,T\}} \sum_{k} \left[\ln S_c(f_k) + \frac{2\Delta f \; |\Delta t \, [d_{c,k} - h_c}{S_c(f_k)} \right. \tag{38}$$

where $d_{c,k}$ and $h_{c,k}$ are the complex local Fourier modes and $S_c(f)$ is the stationary effective PSD, namely instrumental noise plus the time-averaged galactic foreground contribution in channel c. The WDM analysis starts from the same local Fourier patch, embeds the template into that patch, and applies the WDM transform with $N_t = 32$, $a = 1/3$, and $d = 1$. The resulting WDM likelihood is the diagonal Gaussian specialization of Eq. (31),

$$\ln \mathcal{L}_{\mathrm{WDM}}(\theta) = -\frac{1}{2} \sum_{c \in \{A,E,T\}} \sum_{n,m} \left[\ln(2\pi\sigma_{c,m}^2) + \frac{\left(w_{c,nm}^{(d)} - w_{c,nm}^{(h)}(\theta) \right)^2}{\sigma_{c,m}^2} \right], \tag{39}$$

with a stationary per-pixel variance

$$\sigma_{c,m}^2 = \frac{S_c(f_m)}{2\Delta t} = f_{\mathrm{Nyq}} S_c(f_m), \tag{40}$$

broadcast across the WDM time bins. In other words, this example uses the same stationary PSD in both domains and asks whether the WDM transform reproduces the frequency-domain posterior under that shared noise model.

Both analyses use the NumPyro implementation of the No-U-Turn Sampler with a dense mass matrix, target acceptance probability 0.85, and maximum tree depth 10. Each run uses 800 warmup steps followed by 1000 retained samples from a single chain, initialized near the injected parameters with small jitters in $\log \dot{f}$ and $\log A$. In the realization shown here, both samplers produced zero divergences. The latent effective sample sizes are 613–637 for the frequency-domain chain and 590–714 for the WDM chain, while the corresponding $\hat{R}$ values lie in the ranges 0.999–1.007 and 0.999–1.016. The truth-band optimal SNRs are 22.5469 in the frequency-domain calculation and 22.5462 in the WDM calculation. All reported 90% credible intervals contain the injected truth in both runs, and the posterior medians differ only by 9.0×10^{-11} Hz in f_0, $2.2 \times 10^{-20}\,\mathrm{Hz\,s^{-1}}$ in $\dot{f}$, 5.2×10^{-26} in A, 6.1×10^{-3} rad in ϕ_0, and 0.10 in SNR. Within the numerical precision of this study, the WDM and frequency-domain analyses are therefore statistically indistinguishable for this seeded resolved-GB example.

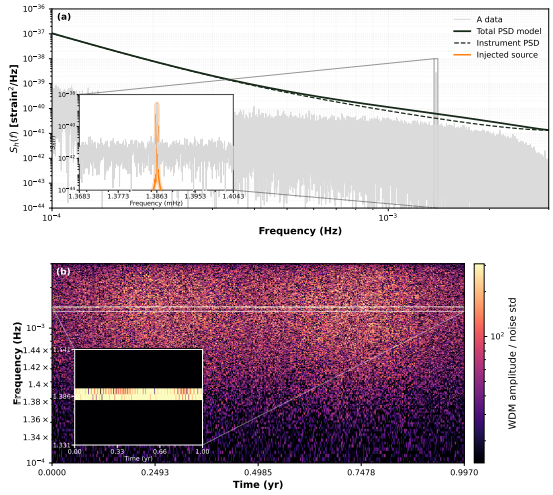

Figure 5. **LISA galactic background: Strain comparison.** Characteristic-strain comparison between the injected resolved galactic binary and the channel noise model including the galactic foreground.

Table I. **Jensen–Shannon divergence between WDM and frequency-domain marginal posteriors across 100 independent noise realisations.** The injected signal parameters are held fixed; only the noise draw varies. JSD is computed on the 1D marginal for each sampled parameter. All values in nats; JSD ≤ 0.01 indicates posteriors that are effectively identical.

Parameter	Median	$[q_{05},\ q_{95}]$	Max
f_0 [Hz]	0.0139	[0.0086, 0.0469]	0.109
$\dot{f}$ [Hz/s]	0.0106	[0.0069, 0.0139]	0.016
A	0.0125	[0.0087, 0.0227]	0.059
ϕ_0 [rad]	0.0153	[0.0051, 0.0686]	0.184

The `PyWavelet` package enables both a straightforward NumPy-based development path and high-performance accelerator execution via JAX. For small data lengths ($N \lesssim 2^{20}$), the NumPy version is competitive or even faster due to its extremely low overhead. However, once N exceeds $\sim 2^{20}$, the GPU-accelerated versions achieve markedly lower runtimes while both backends maintain round-trip reconstruction errors at floating-point roundoff. This combination of flexibility, accuracy, and performance makes `PyWavelet` a useful tool for time-frequency-WDM analysis of long-duration gravitational-wave signals.

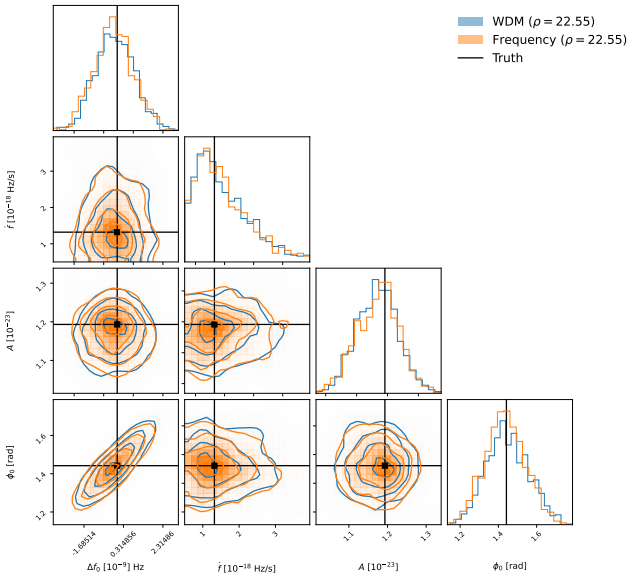

Figure 6. **LISA galactic background: Posterior analysis.** Overlaid posterior corner plot for the WDM-domain and frequency-domain analyses of the same realization. The two posterior families overlap closely in $(f_0, \dot{f}, A, \phi_0)$ and in the derived optimal SNR.

IV. CONCLUSIONS

We have presented a self-contained, pedagogical account of the Wilson-Daubechies-Meyer (WDM) wavelet packet transform and its implementation in the open-source `PyWavelet` package. The main goal of this work is to make the WDM representation accessible to readers who are new to time-frequency methods, by walking through the full three-case basis definition (DC, interior, and Nyquist channels), the packed $(N_t, N_f{+}1)$ coefficient layout, the forward and inverse transforms with detailed derivations, and the conditions under which WDM coefficients can be treated as approximately independent.

On the implementation side, the NumPy and JAX backends both achieve exact round-trip reconstruction to floating-point precision. The JAX backend provides substantially lower runtimes for large transforms, making it practical to incorporate WDM-domain operations into inference pipelines that require millions of likelihood evaluations.

Several limitations of the current work should be noted. First, the package implements only the $d = 1$ Meyer window; higher-order windows ($d > 1$) offer better spectral leakage suppression and may be preferable for data with sharp spectral features. Second, this paper does not address the generation of gravitational-wave signals directly in the WDM domain, which is where much of the computational advantage of the WDM representation lies [1]; we defer fast wavelet-domain waveform generation to future work. Third, the treatment of data gaps and edge effects, while straightforward in principle, requires careful handling of the boundary wavelets and is not yet implemented in the package.

Looking ahead, the most immediate extensions are the incorporation of fast wavelet-domain waveform generation for compact binary signals, the implementation of evolutionary power spectral density estimation directly on the WDM grid, and integration of the package into existing LISA data analysis frameworks. The combination of a pedagogical reference, a tested open-source implementation, and GPU-accelerated performance should lower the barrier for the gravitational-wave community to adopt time-frequency methods for the long-duration, non-stationary signals expected from future detectors.

DATA AND SOFTWARE AVAILABILITY

The software developed for this research is available on GitHub at `https://github.com/pywavelet/wdm_transform`. Documentation and worked examples are available in the repository documentation at `https://github.com/pywavelet/wdm_transform/tree/main/docs`.

ACKNOWLEDGMENTS

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Appendix A: Normality condition of the $g_{nm}(f)$ basis

We introduce the basis $\tilde{g}_{nm}(f)$ in the discrete case:

$$\tilde{g}_{nm}[l] = \frac{1}{\sqrt{2}} \begin{cases} e^{-4\pi i n l/N_t}\, \tilde{\varphi}[l], & m = 0, \\ e^{-2\pi i n l/N_t} \Big(C_{nm}\, \tilde{\varphi}[l - mN_t/2] + C^*_{nm}\, \tilde{\varphi}[l + mN_t/2] \Big), & 0 < m < N_f, \\ e^{-4\pi i n l/N_t} \Big(\tilde{\varphi}[l - N/2] + \tilde{\varphi}[l + N/2] \Big), & m = N_f, \end{cases} \tag{A1}$$

Where we write the Meyer window function in the discrete case focusing on $d = 1$:

$$\tilde{\varphi}[l] = \sqrt{\frac{2}{N_t}} \begin{cases} 1 & |l_r| < A \\ \cos\left[\frac{\pi}{2}\left(\frac{|l_r|-A}{B}\right)\right] & A \leq |l_r| < A+B \\ 0 & |l_r| \geq A+B \end{cases}$$

$$l_r = l\frac{2}{N_t} \tag{A2}$$

where $l \in \{-\frac{N_t}{2}, -\frac{N_t}{2}+1, ..., \frac{N_t}{2}-1\}$. Now we verify that

$$\begin{aligned}\sum_{nm} \tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] &= \frac{1}{2}\sum_n e^{-4\pi i n(l-l')/N_t}\,\tilde{\varphi}[l]\,\tilde{\varphi}[l'] \\ &+ \frac{1}{2}\sum_{0<m<N_f}\sum_n e^{-2\pi i n(l-l')/N_t}\Big(C_{nm}\,\tilde{\varphi}[l-mN_t/2] + C^*_{nm}\,\tilde{\varphi}[l+mN_t/2]\Big)\Big(C^*_{nm}\,\tilde{\varphi}[l'-mN_t/2] + C_{nm}\,\tilde{\varphi}[l'+mN_t/2]\Big) \\ &+ \frac{1}{2}\sum_n e^{-4\pi i n(l-l')/N_t}\Big(\tilde{\varphi}[l-N/2] + \tilde{\varphi}[l+N/2]\Big)\Big(\tilde{\varphi}[l'-N/2] + \tilde{\varphi}[l'+N/2]\Big)\end{aligned} \tag{A3}$$

Now we simplify the $0 < m < N_f$ term, by noticing that $C^2_{nm} = C^{*2}_{nm} = (-1)^m(-1)^n$ and $|C_{nm}|^2 = 1$

$$\begin{aligned}\frac{1}{2}\sum_{0<m<N_f}\sum_n e^{-2\pi i n(l-l')/N_t}\Big(&\tilde{\varphi}[l-mN_t/2]\tilde{\varphi}[l'-mN_t/2] + \tilde{\varphi}[l+mN_t/2]\tilde{\varphi}[l'+mN_t/2] \\ &+ (-1)^m(-1)^n\tilde{\varphi}[l-mN_t/2]\tilde{\varphi}[l'+mN_t/2] + (-1)^m(-1)^n\tilde{\varphi}[l+mN_t/2]\tilde{\varphi}[l'-mN_t/2]\Big)\end{aligned} \tag{A4}$$

one notices that the summation over n can be directly performed, and it vanishes for every value but $l = l'$. In particular, the second line (the mixed terms) always vanishes. A similar result holds for the $m = 0, N_f$ cases, therefore we can write

$$\sum_{nm}\tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] = \delta_{ll'}\frac{N_t}{2}\left(\tilde{\varphi}^2[l] + \sum_{0<m<N_f}\tilde{\varphi}^2[l-mN_t/2] + \sum_{0<m<N_f}\tilde{\varphi}^2[l+mN_t/2] + \tilde{\varphi}^2[l-N/2] + \tilde{\varphi}^2[l+N/2]\right) \tag{A5}$$

Now the thing to notice is that each value of l lays in two and only two of these windows. Let be $\bar{m}$ and $\bar{m}+1$ the indices of these two windows, such that $\frac{\bar{m}N_t}{2} < |l| < \frac{(\bar{m}+1)N_t}{2}$ ($\bar{m}$ could be 0, or $\bar{m}+1$ can be N_f). Then the sum becomes

$$\sum_{nm}\tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] = \delta_{ll'}\frac{N_t}{2}\left(\tilde{\varphi}^2[l-\bar{m}N_t/2] + \tilde{\varphi}^2[l-(\bar{m}+1)N_t/2]\right) \tag{A6}$$

Let $l \geq 0$ for sake of simplicity ($l < 0$ cases are analogous). Then either $l < \frac{(m+A)N_t}{2}$ or $l \geq \frac{(m+A)N_t}{2}$. In the first case, we have $\tilde{\varphi}^2[l-\bar{m}N_t/2] = \frac{2}{N_t}$ and $\tilde{\varphi}^2[l-(\bar{m}+1)N_t/2] = 0$, since for $|\frac{2l}{N_t} - (\bar{m}+1)| > A+B$. In the second case, we have

$$\sum_{nm}\tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] = \delta_{ll'}\left(\cos^2\left[\frac{\pi}{2}\frac{\frac{2l}{N_t}-\bar{m}-A}{B}\right] + \cos^2\left[\frac{\pi}{2}\frac{\bar{m}+1-\frac{2l}{N_t}-A}{B}\right]\right) \tag{A7}$$

which after some algebra simplifies to

$$\sum_{nm}\tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] = \delta_{ll'}\left(1 + \cos\left(\frac{\pi}{2}\frac{1-2A}{B}\right)\cos\left(2\frac{\pi}{2}\frac{\frac{2l}{N_t}-\bar{m}-A}{B} - \frac{\pi}{2}\frac{1-2A}{B}\right)\right) = \delta_{ll'} \tag{A8}$$

where we used the fact of $2A + B = 1$. Therefore we find that for every value of l we have

$$\sum_{nm}\tilde{g}_{nm}[l]\tilde{g}^*_{nm}[l'] = \delta_{ll'} \tag{A9}$$

1. Normalization of the Meyer window

We verify that the Meyer window defined in Eq. (A2) (the discrete case of (15)) is correctly normalized. We compute

$$\sum_{l=-N/2}^{N/2-1} \tilde{\varphi}^2[l] = 2 \sum_{l=0}^{A\frac{N_t}{2}-1} \frac{2}{N_t} - \frac{2}{N_t} + 2 \sum_{l=A\frac{N_t}{2}}^{(A+B)\frac{N_t}{2}-1} \frac{2}{N_t} \cos^2\left[\frac{\pi}{2}\frac{\frac{2l}{N_t}-A}{B}\right] = 2A - \frac{2}{N_t} + \frac{4}{N_t} \sum_{k=0}^{B\frac{N_t}{2}-1} \cos^2\left[\frac{\pi}{2}\frac{2k}{BN_t}\right] =$$

$$= 2A - \frac{2}{N_t} + \frac{4}{N_t} \sum_{k=0}^{B\frac{N_t}{2}-1} \frac{1}{2} + \frac{4}{N_t} \sum_{k=0}^{B\frac{N_t}{2}-1} \frac{1}{2} \cos\left(\frac{2\pi k}{BN_t}\right) = 2A - \frac{2}{N_t} + B + \frac{2}{N_t} = 1 \tag{A10}$$

where on the first line the term $-\frac{2}{N_t}$ avoids double counting $l = 0$, where the last summation equals $+\frac{2}{N_t}$ after performing the geometrical series and where we remember that $2A + B = 1$ is a property of the window function.

Appendix B: Detailed derivation for the forward and inverse transforms

This appendix provides the full algebraic steps leading to the compact expressions for the forward WDM coefficients presented in Section II. We work in the Fourier domain throughout, writing the forward transform as

$$w_{nm} = \sum_{l=0}^{N-1} \tilde{x}_l \, \tilde{g}^*_{nm}(f[l]) \,, \tag{B1}$$

where the discrete frequencies are mapped as

$$f[l] = \begin{cases} l & 0 \le l < N/2 \\ l - N & N/2 \le l < N \end{cases} \times df \,, \qquad df = \frac{1}{T} \,. \tag{B2}$$

1. Interior channels ($0 < m < N_f$)

Inserting the interior-channel basis element from Eq. (14) and splitting the sum over positive and negative frequencies gives

$$\begin{aligned} w_{nm} = {} & \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l \, e^{2\pi i l n/N_t} \left[C^*_{nm} \tilde{\varphi}\big((l - \tfrac{mN_t}{2})df\big) + C_{nm} \tilde{\varphi}\big((l + \tfrac{mN_t}{2})df\big)\right] \\ & + \frac{1}{\sqrt{2}} \sum_{l=N/2+1}^{N-1} \tilde{x}_l \, e^{2\pi i (l-N) n/N_t} \left[C^*_{nm} \tilde{\varphi}\big((l-N - \tfrac{mN_t}{2})df\big) + C_{nm} \tilde{\varphi}\big((l-N + \tfrac{mN_t}{2})df\big)\right] . \end{aligned} \tag{B3}$$

Substituting $l \to N - l$ in the second sum and using the reality condition $\tilde{x}_{N-l} = \tilde{x}^*_l$ together with the symmetry $\tilde{\varphi}(-f) = \tilde{\varphi}(f)$, the second sum is the complex conjugate of the first. The boundary terms at $l = 0$ and $l = N/2$ vanish because the window support does not extend to those frequencies for interior channels. We therefore obtain

$$\begin{aligned} w_{nm} &= \sqrt{2}\, \Re \sum_{l=1}^{N/2-1} \tilde{x}_l \, e^{2\pi i l n/N_t} \, C^*_{nm} \, \tilde{\varphi}\big((l - \tfrac{mN_t}{2})df\big) \\ &= \sqrt{2}\, (-1)^{nm} \, \Re \sum_{l'=-N_t/2}^{N_t/2} \tilde{x}_{l'+mN_t/2} \, e^{2\pi i l' n/N_t} \, C^*_{nm} \, \tilde{\varphi}(l' \, df) \,, \end{aligned} \tag{B4}$$

where the last equality follows from the substitution $l' = l - mN_t/2$ and the fact that $\tilde{\varphi}(l' \, df)$ is nonzero only for $|l'| \le N_t/2$. This is Eq. (21) of the main text, and corresponds (up to normalization conventions) to Eq. 17 of Cornish [1].

2. DC channel ($m = 0$)

For the DC edge channel, the basis element has a doubled time-shift exponent and no C_{nm} factor. The same positive/negative frequency splitting yields

$$\begin{aligned} w_{n0} &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l \, e^{4\pi i l n/N_t} \, \tilde{\varphi}(l\, df) + \frac{1}{\sqrt{2}} \sum_{l=N/2}^{N-1} \tilde{x}_l \, e^{4\pi i (l-N) n/N_t} \, \tilde{\varphi}((l-N)\, df) \\ &= \sqrt{2} \Re \left[\sum_{l=1}^{N_t/2} \tilde{x}_l \, e^{4\pi i l n/N_t} \, \tilde{\varphi}(l\, df) \right] + \frac{1}{\sqrt{2}} \tilde{x}_0 \, \tilde{\varphi}(0) \,, \end{aligned} \tag{B5}$$

which is Eq. (24) of the main text.

3. Nyquist channel ($m = N_f$)

$$\begin{aligned} w_{nm} &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l \, e^{2\pi i l n/N_t} \left[\tilde{\varphi}\big((l - \tfrac{N}{2}) df\big) + \tilde{\varphi}\big((l + \tfrac{N}{2}) df\big) \right] + \frac{1}{\sqrt{2}} \sum_{l=N/2}^{N-1} \tilde{x}_l \, e^{2\pi i (l-N) n/N_t} \left[\tilde{\varphi}\big((l - N - \tfrac{N}{2}) df\big) + \tilde{\varphi}\big((l - N + \tfrac{N}{2}) df\big) \right] \\ &= \frac{1}{\sqrt{2}} \sum_{l=0}^{N/2-1} \tilde{x}_l \, e^{2\pi i l n/N_t} \left[\tilde{\varphi}\big((l - \tfrac{N}{2}) df\big) + \tilde{\varphi}\big((l + \tfrac{N}{2}) df\big) \right] + \frac{1}{\sqrt{2}} \sum_{l=1}^{N/2} \tilde{x}_{N-l} \, e^{-2\pi i l n/N_t} \left[\tilde{\varphi}\big((-l - \tfrac{N}{2}) df\big) + \tilde{\varphi}\big((-l + \tfrac{N}{2}) df\big) \right] . \end{aligned} \tag{B6}$$

the $l = 0$ term in the first summation vanishes, and the $l = N/2$ term is treated separately:

$$\begin{aligned} w_{nm} &= \sqrt{2}\, \Re \sum_{l=1}^{N/2-1} \tilde{x}_l \, e^{2\pi i l n/N_t} \, \tilde{\varphi}\big((l - \tfrac{N}{2}) df\big) + \frac{1}{\sqrt{2}} \tilde{x}_{N/2} \, e^{-\pi i n N_f} \, \tilde{\varphi}(0) \\ &= \sqrt{2}\, \Re \sum_{l'=-N_t/2}^{-1} \tilde{x}_{l'+N/2} \, e^{2\pi i l' n/N_t} \, \tilde{\varphi}(l'\, df) + \frac{1}{\sqrt{2}} \tilde{x}_{N/2} \, \tilde{\varphi}(0) \,, \end{aligned} \tag{B7}$$

assuming an even value for N_f.

4. The inverse transform

We write the inverse transform of the w_{nm} coefficients to recover the signal $x[l]$ for $l < N/2$ since we know that for $l \geq N/2$ the reality condition holds ($x[N-l] = x^*[l]$):

$$\begin{aligned} x[l] = \sum_{n=0}^{N_t-1} \sum_{m=0}^{N_f} w_{nm} g_{nm}[l] &= \frac{1}{\sqrt{2}} \sum_n \sum_{0<m<N_f} w_{nm} e^{-2\pi i l n/N_t} \left[C_{nm} \tilde{\varphi}\big((l - \tfrac{mN_t}{2}) df\big) + C^*_{nm} \tilde{\varphi}\big((l + \tfrac{mN_t}{2}) df\big) \right] \\ &\quad + \frac{1}{\sqrt{2}} \sum_n w_{n0} e^{-4\pi i l n/N_t} \tilde{\varphi}(l\, df) + \frac{1}{\sqrt{2}} \sum_n w_{nN_f} e^{-4\pi i l n/N_t} \left[\tilde{\varphi}\big((l - \tfrac{N}{2}) df\big) + \tilde{\varphi}\big((l + \tfrac{N}{2}) df\big) \right] = \\ &= \begin{cases} x_0[l] & l < N_t/2 \\ x_m[l] & mN_t/2 \leq l < (m+1)N_t/2 \\ x_{N_f}[l] & |l| \geq N/2 - N_t/2 \end{cases} \end{aligned} \tag{B8}$$

where we note that for each vlaue of l, the inverse transform depends on which of the m bands the l value belongs, in particular

$$
\begin{aligned}
x_0[l] &= \frac{1}{\sqrt{2}}\sum_n w_{n1} e^{-2\pi i l n/N_t} C_{n1}\tilde{\varphi}\big((l-\tfrac{N_t}{2})df\big) + \sum_n w_{n0} e^{-4\pi i l n/N_t}\tilde{\varphi}(l\,df) \\
x_m[l] &= \frac{1}{\sqrt{2}}\sum_n w_{nm} e^{-2\pi i l n/N_t} C_{nm}\tilde{\varphi}\big((l-\tfrac{mN_t}{2})df\big) + \frac{1}{\sqrt{2}}\sum_n w_{n(m+1)} e^{-2\pi i l n/N_t} C_{n(m+1)}\tilde{\varphi}\Big((l-\tfrac{(m+1)N_t}{2})df\Big) \\
x_m[l] &= \frac{1}{\sqrt{2}}\sum_n w_{n(N_f-1)} e^{-2\pi i l n/N_t} C_{n(N_f-1)}\tilde{\varphi}\big((l-\tfrac{N}{2}+\tfrac{N_t}{2})df\big) + \frac{1}{\sqrt{2}}\sum_n w_{nN_f} e^{-4\pi i l n/N_t}\tilde{\varphi}\big((l-\tfrac{N}{2})df\big) \qquad \text{(B9)}
\end{aligned}
$$

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